Abdul Khurram

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Education & Awards

University of British Columbia — Vancouver, BC, Canada
*Bachelor of Science in Computer Science (**Expected Graduation: May 2027**)*

Euclid Mathematics Contest, University of Waterloo — 2021
Top 5% Internationally, Certificate of Distinction

Work Experience

Incoming Software Engineer Intern — Sept. – Dec. 2026
Shopify — *Toronto, ON, Canada*

Software Developer Co-op | *Python, Scala, **Kafka**, **Airflow*** — Jan. 2026 – Present
RBC Royal Bank — *Vancouver, BC, Canada*

- Engineered **Python/Scala data pipelines** ingesting **50M+ daily events** via Kafka and Airflow, enforcing RBAC/encryption on **petabyte-scale Delta Lake**.
- Optimized **Apache Spark** batch jobs via Spark-native API migration and AQE tuning, cutting latency by **40%**; built dashboards tracking pipeline health.
- Shipping multi-region services on **Kubernetes**-orchestrated clusters with fault tolerance, idempotent re-runs, and **Terraform**-provisioned failover.

Software Engineer Intern | ***Python**, TypeScript/Node.js, **AWS**, Docker* — Jan. – Dec. 2025
Contello.ai — *Toronto, ON, Canada*

- Built async **Python** data pipelines on **AWS** (SQS, S3, VPC) processing **thousands of jobs/day** with dead-letter routing and back-pressure controls.
- Raised service reliability to **99.9%** via retry/backoff, Elasticsearch-indexed logs, CloudWatch alerting, and structured observability across distributed services.

Research Experience

Software Developer & Team Lead | ***C++**, **Python**, Unreal Engine* — Sept. – Dec. 2024
UBC Emerging Media Lab — *Vancouver, BC, Canada*

- Led development of a real-time VR simulation in **C++**, designing a data collection pipeline streaming sensor events for AI decision-making.
- Built a multithreaded supervisory system (**C++/Python**) coordinating concurrent actors with shared state and lock-free message passing.
- Profiled the processing pipeline, identified memory and I/O hotspots, and restructured the call graph—cutting latency by **40%** and costs by **90%**.

Projects

Backer | ***Go**, **Python**, **AWS** (**DynamoDB**, S3), XGBoost* — **Hack The Coast 2026**

- Designed a **DynamoDB**-backed data layer (10+ tables) with batch operations, transactional writes, and typed abstractions for high-throughput workflows.
- Built a **Go** data ingestion service consuming **DynamoDB** event streams; engineered features and trained an XGBoost ranking model scoring entities on 6 signals.

Workbase | *TypeScript, **PostgreSQL**, **AWS** Bedrock, Vitest* — **Personal Project**

- Built a career content platform that ingests GitHub repos via OAuth, normalizes commit/PR data into **PostgreSQL**, and generates verified highlights with evidence attribution.
- Designed a multi-step **data pipeline**: LLM generation, schema validation, structured output repair, audit logging—with deterministic retry and automated tests.

Skills

Languages: C/C++, Python, Go, Scala, Java, SQL, TypeScript/JavaScript
Infrastructure: Kubernetes, Docker, Terraform, Kafka, Airflow, Apache Spark, AWS (S3, SQS, Lambda, DynamoDB, CloudWatch); Linux, Git, CI/CD
Databases & Storage: PostgreSQL, DynamoDB, Elasticsearch, Redis, Delta Lake; REST APIs, Flask/FastAPI